

JEE MAIN REVISION MASTER: SOME BASIC CONCEPTS OF CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Mole Concept, Molar Mass & Stoichiometry	13 PYQs	Quantitative Stoichiometry
Standard	Concentration Terms: Molarity, Molality & Mole Fraction	2 PYQs	Solutions & Concentrations

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mole Concept, Molar Mass & Stoichiometry	$n = \frac{w}{M} = \frac{N}{6.022 \times 10^{23}} = \frac{V_{\text{at STP}}}{22.4 \text{ L}}$ $\text{Empirical Formula} = (\text{Simplest Ratio})_n$
Concentration Terms: Molarity, Molality & Mole Fraction	$M = \frac{w_B \times 1000}{M_B \times V_{\text{mL}}}, \quad m = \frac{w_B \times 1000}{M_B \times w_A(\text{g})}, \quad M_1 V_1 = M_2 V_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mole Concept, Molar Mass & Stoichiometry (High Yield — 13 PYQs)

Core Principle: Mole concept calculations: $n = (w / M) = (N / N_A) = (V_{\text{at STP}} / 22.4 \text{ L})$. Stoichiometry and identification of limiting reagents that determine product yield.

■ **Caution / JEE Pitfall:** At STP (1 bar, 273.15 K), molar volume of ideal gas is 22.7 L; at standard old STP (1 atm, 0°C), it is 22.4 L.

■ **Speed Trick:** Number of atoms in a sample = Moles $\times N_A \times$ Atomicity.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Match List-I with List-II and select the correct option (Based on mole concept): List-I List-II (a) 2 moles of ethene (i) 11.2 L volume ..." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "From the given information, select the suitable law of chemical combination: Cupric Carbonate % of Cu % of C % of O Natural Sample 51.35 9.7..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "Select the INCORRECT statement/s from the following: (a) 22 books have infinite significant figures (b) In the answer of calculation $\$2.5 \text{ L} \dots$ " **[Ans: N/A]**

3.2 Concentration Terms: Molarity, Molality & Mole Fraction (Standard Yield — 2 PYQs)

Core Principle: Molarity $M = (\text{moles of solute} / \text{Volume of solution (L)})$ (temperature dependent). Molality $m = (\text{moles of solute} / \text{Mass of solvent (kg)})$ (temperature independent). Mole fraction $\chi_B = (n_B / n_A + n_B)$.

■ **Caution / JEE Pitfall:** Molality and mole fraction are independent of temperature because they depend strictly on masses, not solution volume.

■ **Speed Trick:** Relation between Molarity (M) and Molality (m): $m = (1000 M / 1000 d - M M_B)$ where d is density in g/mL.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "Which of the following methods of expressing concentration are unitless?..." [Ans: A]
- [2022 Shift] Q#2 (MCQ): "An aqueous solution of alcohol contains 18g of water and 414g of ethyl alcohol. The mole fraction of water is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II and select the correct option (Based on mole concept): List-I List-II (a) 2 moles of ethene (i) 11.2 L volume at STP (b) Mass is equal to 66 g (ii) 56 g (c) 1 g of H_2 (iii) 12.04×10^{23} Molecules (d) 2 moles of water vapours (iv) 1.5 mole of CO_2

- (A) a-ii, b-iv, c-i, d-iii
- (B) a-iii, b-i, c-iv, d-ii
- (C) a-i, b-iv, c-ii, d-iii
- (D) a-ii, b-iii, c-i, d-iv

Correct Answer: None

Solution Analysis: For mole concept, we use these basic relations: 1 mole of any substance contains 6.02×10^{23} particles. 1 mole of any gas at STP occupies 22.4 L. Also, Mass=Number of moles $\times$ Molar mass For (a), ethene is C_2H_4 . Its molar mass is $2 \times 12 + 4 \times 1 = 28 \text{ g mol}^{-1}$ So, mass of 2 moles of ethene is $2 \times 28 = 56 \text{ g}$ Hence, (a) $\rightarrow$ (ii) For (b), mass is 66 g. This matches carbon dioxide, CO_2 , because molar mass of CO_2 is $12 + 2 \times 16 = 44 \text{ g mol}^{-1}$ Number of moles of

Benchmark Problem 2: KCET 2025 (Q#3)

Which of the following methods of expressing concentration are unitless?

- (A) Mole fraction and Mass percent (W/W) [Correct]
- (B) Molality and Mole fraction
- (C) Mass percent (W/W) and Molality
- (D) Molality and Molarity

Correct Answer: A

Solution Analysis: Option A is correct: mole fraction and mass percent (w/w) are both unitless. Mole fraction $x_i = \frac{n_i}{\sum n_j}$ It's simply a ratio of moles, so no units. Mass percent (w/w) mass % = $(\text{mass of solute} / \text{total mass of solution}) \times 100\%$ Again, a ratio of masses, giving a percentage without units. By contrast, molality has units of mol/kg and molarity has mol/L, so neither is unitless.